

Speed, Distance & Time

Answer Key & Detailed Solutions — Questions 1 to 21

Question 1

A boat takes 2 hours to travel downstream a river from port A to port B, and 3 hours to return to port A. Another boat takes a total of 6 hours to travel from port B to port A and return to port B. If the speeds of the boats and the river are constant, then the time, in hours, taken by the slower boat to travel from port A to port B is

A) $12(\sqrt{5} - 2)$

B) $3(3 + \sqrt{5})$

C) $3(\sqrt{5} - 1)$

✓ D) $3(3 - \sqrt{5})$ ← Correct Answer

Solution:

Let boat 1 speed = b_1 , boat 2 speed = b_2 , river speed = r , distance A to B = d .

Boat 1:

Downstream: $d/(b_1 + r) = 2 \rightarrow b_1 + r = d/2$

Upstream: $d/(b_1 - r) = 3 \rightarrow b_1 - r = d/3$

Adding: $2b_1 = 5d/6 \rightarrow b_1 = 5d/12$. Subtracting: $2r = d/6 \rightarrow r = d/12$.

Boat 2 (total round trip = 6 hours):

$$d/(b_2 - r) + d/(b_2 + r) = 6$$

$$2b_2 \cdot d / (b_2^2 - r^2) = 6$$

Substitute $r = d/12$: $2b_2 \cdot d / (b_2^2 - d^2/144) = 6$

Divide by d^2 : $(b_2/d) = (2 + \sqrt{5})/12$ [taking positive root of quadratic $3X^2 - X - 1/48 = 0$ where $X = b_2/d$].

Compare: $b_1/d = 5/12 \approx 0.417$; $b_2/d = (2 + \sqrt{5})/12 \approx 0.353$. Since $b_1 > b_2$, Boat 2 is slower.

Time for Boat 2 (slower) downstream A to B:

$$\text{Time} = d/(b_2 + r) = 1/[(2 + \sqrt{5})/12 + 1/12] = 12/(3 + \sqrt{5}).$$

Rationalise: $12(3 - \sqrt{5})/[(3 + \sqrt{5})(3 - \sqrt{5})] = 12(3 - \sqrt{5})/(9 - 5) = 12(3 - \sqrt{5})/4 = 3(3 - \sqrt{5})$.

Question 2

Points A, P, Q and B lie on the same line such that P, Q and B are, respectively, 100 km, 200 km and 300 km away from A. Cars 1 and 2 leave A at the same time and move towards B. Simultaneously, car 3 leaves B and moves towards A. Car 3 meets car 1 at Q, and car 2 at P. If each car is moving in uniform speed then the ratio of the speed of car 2 to that of car 1 is

A) 2:7

B) 2:9

C) 1:2

✓ D) 1:4 ← Correct Answer

Solution:

Positions: A=0, P=100, Q=200, B=300 km.

Car 1 and Car 3 meet at Q (200 km from A):

Car 1 travels 200 km from A. Car 3 travels $300 - 200 = 100$ km from B.

Same time: $200/s_1 = 100/s_3 \rightarrow s_1 = 2 \cdot s_3$.

Car 2 and Car 3 meet at P (100 km from A):

Car 2 travels 100 km from A. Car 3 travels $300 - 100 = 200$ km from B.

Same time: $100/s_2 = 200/s_3 \rightarrow s_2 = s_3/2$.

Ratio $s_2/s_1 = (s_3/2)/(2 \cdot s_3) = 1/4$.

The ratio of speed of car 2 to car 1 is 1:4.

Question 3

Points A and B are 150 km apart. Cars 1 and 2 travel from A to B, but car 2 starts from A when car 1 is already 20 km away from A. Each car travels at a speed of 100 kmph for the first 50 km, at 50 kmph for the next 50 km, and at 25 kmph for the last 50 km. The distance, in km, between car 2 and B when car 1 reaches B is

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Solution:

Total time for Car 1 to travel 150 km:

Segment 1 (50 km at 100 kmph): 0.5 hr

Segment 2 (50 km at 50 kmph): 1.0 hr

Segment 3 (50 km at 25 kmph): 2.0 hr

Total = 3.5 hours.

Car 2 starts when Car 1 is 20 km from A. Car 1 covers 20 km at 100 kmph in 0.2 hrs.

So Car 2 starts 0.2 hrs after Car 1.

When Car 1 reaches B (3.5 hrs after Car 1 starts), Car 2 has been travelling $3.5 - 0.2 = 3.3$ hours.

Distance covered by Car 2 in 3.3 hours:

First 100 km (seg 1 + seg 2) takes $0.5 + 1.0 = 1.5$ hours.

Remaining time = $3.3 - 1.5 = 1.8$ hours in segment 3 (25 kmph).

Distance in segment 3 = $25 \times 1.8 = 45$ km.

Total distance by Car 2 = $50 + 50 + 45 = 145$ km.

Distance from Car 2 to B = $150 - 145 = 5$ km.

Question 4

On a long stretch of east-west road, A and B are two points such that B is 350 km west of A. One car starts from A and another from B at the same time. If they move towards each other, then they meet after 1 hour. If they both move towards east, then they meet in 7 hrs. The difference between their speeds, in km per hour, is

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Solution:

Let speeds be s_A (from A) and s_B (from B).

Case 1 (towards each other): $s_A + s_B = 350/1 = 350 \dots(1)$

Case 2 (both east, $s_A > s_B$ as A catches B):

$350 = (s_A - s_B) \times 7$

$s_A - s_B = 50 \dots(2)$

The difference between their speeds = 50 km/hr.

Question 5

Two trains cross each other in 14 seconds when running in opposite directions along parallel tracks. The faster train is 160 m long and crosses a lamp post in 12 seconds. If the speed of the other train is 6 km/hr less than the faster one, its length, in m, is

- 184) 184
- 192) 192
- 190) 190
- 180) 180

Solution:

Speed of faster train = $160 \text{ m} / 12 \text{ s} = 40/3 \text{ m/s} = 48 \text{ km/hr}$.

Speed of slower train = $48 - 6 = 42 \text{ km/hr} = 42 \times (5/18) = 35/3 \text{ m/s}$.

Relative speed (opposite directions) = $40/3 + 35/3 = 75/3 = 25 \text{ m/s}$.

When crossing each other: total distance = sum of lengths = $(160 + L)$.

Time = 14 s.

$160 + L = 25 \times 14 = 350$.

$L = 350 - 160 = 190 \text{ m}$.

Question 6

A train travelled at one-third of its usual speed, and hence reached the destination 30 minutes after the scheduled time. On its return journey, the train initially travelled at its usual speed for 5 minutes but then stopped for 4 minutes for an emergency. The percentage by which the train must now increase its usual speed so as to reach the destination at the scheduled time is nearest to:

- A) 50
- B) 58
- ✓ C) 67 ← Correct Answer
- D) 61

Solution:

Let usual speed = S , scheduled time = T , distance = $D = S \cdot T$.

At $S/3$: time = $3T$. Delay = $3T - T = 2T = 30 \text{ min} \rightarrow T = 15 \text{ min} = 0.25 \text{ hr}$.

$D = S \times 0.25 = S/4$.

Return journey:

Travels 5 min at speed S : distance = $S \times (5/60) = S/12$.

Remaining distance = $S/4 - S/12 = 3S/12 - S/12 = 2S/12 = S/6$.

Time elapsed = $5 + 4 = 9 \text{ min}$. Time remaining = $15 - 9 = 6 \text{ min} = 1/10 \text{ hr}$.

Required speed = $(S/6) / (1/10) = 10S/6 = 5S/3$.

Increase = $5S/3 - S = 2S/3$.

% increase = $(2S/3)/S \times 100 = 200/3 \approx 66.67\% \approx 67$.

Question 7

A straight road connects points A and B. Car 1 travels from A to B and Car 2 travels from B to A, both leaving at the same time. After meeting each other, they take 45 minutes and 20 minutes, respectively, to complete their journeys. If Car 1 travels at a speed of 60 km/hr, then the speed of Car 2, in km/hr, is:

- A) 100
- ✓ B) 90 ← Correct Answer
- C) 80
- D) 70

Solution:

Standard formula: $S_1/S_2 = \sqrt{t_2/t_1}$, where t_1 and t_2 are times taken after meeting.

t_1 (Car 1 after meeting) = 45 min = $3/4$ hr.

t_2 (Car 2 after meeting) = 20 min = $1/3$ hr.

$$60/S_2 = \sqrt{(1/3)/(3/4)} = \sqrt{4/9} = 2/3.$$

$$S_2 = 60 \times (3/2) = 90 \text{ km/hr.}$$

Question 8

Two cars travel from different locations at constant speeds. To meet each other after starting at the same time, they take 1.5 hours if they travel towards each other, but 10.5 hours if they travel in the same direction. If the speed of the slower car is 60 km/hr, then the distance traveled, in km, by the slower car when it meets the other car while traveling towards each other, is

- 100) 100
- 90) 90
- 120) 120
- 150) 150

Solution:

Let speeds be S_f (faster) and $S_s = 60$ km/hr (slower). Initial distance = D .

$$\text{Towards each other: } (S_f + S_s) \times 1.5 = D \dots (1)$$

$$\text{Same direction: } (S_f - S_s) \times 10.5 = D \dots (2)$$

$$\text{Equating: } (S_f + 60) \times 1.5 = (S_f - 60) \times 10.5$$

$$1.5S_f + 90 = 10.5S_f - 630$$

$$720 = 9S_f \rightarrow S_f = 80 \text{ km/hr.}$$

Distance by slower car when meeting (towards each other):

$$= S_s \times \text{time} = 60 \times 1.5 = 90 \text{ km.}$$

Question 9

Two persons are walking beside a railway track at respective speeds of 2 and 4 km per hour in the same direction. A train came from behind them and crossed them in 90 and 100 seconds, respectively. The time, in seconds, taken by the train to cross an electric post is nearest to:

- A) 87
- ✓ B) 82 ← Correct Answer
- C) 78
- D) 75

Solution:

Let train length = L km, train speed = S km/hr.

Crossing person 1 (speed 2 km/hr) in 90 s = $1/40$ hr:

$$L = (S - 2) \times (1/40) \dots(1)$$

Crossing person 2 (speed 4 km/hr) in 100 s = $1/36$ hr:

$$L = (S - 4) \times (1/36) \dots(2)$$

$$\text{Equate: } (S-2)/40 = (S-4)/36$$

$$36(S-2) = 40(S-4)$$

$$36S - 72 = 40S - 160$$

$$4S = 88 \rightarrow S = 22 \text{ km/hr.}$$

$$L = (22-2)/40 = 20/40 = 0.5 \text{ km.}$$

$$\text{Time to cross electric post} = L/S = 0.5/22 \text{ hr} = 0.5/22 \times 3600 \text{ s} = 1800/22 = 900/11 \approx 81.8 \text{ s} \approx 82 \text{ s.}$$

Question 10

Moody takes 30 seconds to finish riding an escalator if he walks on it at his normal speed in the same direction. He takes 20 seconds to finish riding the escalator if he walks at twice his normal speed in the same direction. If Moody decides to stand still on the escalator, then the time, in seconds, needed to finish riding the escalator is

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Solution:

Let D = escalator length, v_m = Moody's speed, v_e = escalator speed.

$$\text{Case 1: } D = (v_m + v_e) \times 30 \dots(1)$$

$$\text{Case 2: } D = (2v_m + v_e) \times 20 \dots(2)$$

$$\text{From (1) = (2): } 30(v_m + v_e) = 20(2v_m + v_e)$$

$$30v_m + 30v_e = 40v_m + 20v_e$$

$$10v_e = 10v_m \rightarrow v_m = v_e.$$

$$\text{Substitute into (1): } D = (v_e + v_e) \times 30 = 60v_e.$$

$$\text{Standing still: Time} = D/v_e = 60v_e/v_e = 60 \text{ seconds.}$$

Question 11

A bus starts at 9 am and follows a fixed route. One day, it traveled at a constant speed of 60 km/h and reached its destination 3.5 hours later than its scheduled arrival time. The next day, it traveled two-thirds of its route in one-third of its total scheduled travel time and the remaining part of the route at 40 km/h to arrive exactly on schedule. What is the scheduled arrival time of the bus?

✓ A) 7:30 PM ← Correct Answer

B) 7:00 PM

C) 9:00 PM

D) 10:30 PM

Solution:

Let scheduled travel time = t hours, distance = D km. Scheduled speed $S = D/t$.

Day 1: Speed = 60 km/hr, time = $t + 3.5$ hrs. $D = 60(t + 3.5) \dots(1)$

Day 2: Travels $2D/3$ in $t/3$ hours (speed = $2D/3 \div t/3 = 2D/t = 2S$).

Remaining $D/3$ at 40 km/hr, takes $2t/3$ hours.

Speed for remaining = $(D/3)/(2t/3) = D/(2t) = S/2$.

Given $S/2 = 40 \rightarrow S = 80$ km/hr.

From $D = S \cdot t = 80t$ and equation (1): $80t = 60(t + 3.5)$

$80t = 60t + 210 \rightarrow 20t = 210 \rightarrow t = 10.5$ hours.

Scheduled arrival = 9:00 am + 10.5 hours = 7:30 PM.

Question 12

Leaving home at the same time, Amal reaches the office at 10:15 am if he travels at 8 km/hr, and at 9:40 am if he travels at 15 km/hr. Leaving home at 9:10 am, at what speed, in km/hr, must he travel so as to reach the office exactly at 10:00 am?

A) 13

✓ B) 12 ← Correct Answer

C) 14

D) 11

Solution:

Let D = distance to office, usual departure time = T_{dep} .

At 8 km/hr arrives 10:15 am; at 15 km/hr arrives 9:40 am.

Difference in arrival = 35 min = $7/12$ hr.

$D/8 - D/15 = 7/12$

$(15D - 8D)/120 = 7/12$

$7D/120 = 7/12 \rightarrow D = 10$ km.

Time at 8 km/hr = $10/8 = 1.25$ hr = 1 hr 15 min.

Departure time = 10:15 am - 1:15 = 9:00 am.

New scenario: departs 9:10 am, must arrive 10:00 am.

Travel time = 50 min = $5/6$ hr.

Required speed = $10 / (5/6) = 12$ km/hr.

Question 13

Two ships meet mid-ocean, and then, one ship goes south and the other ship goes west, both travelling at constant speeds. Two hours later, they are 60 km apart. If the speed of one of the ships is 6 km per hour more than the other one, then the speed, in km per hour, of the slower ship is

20) 20

12) 12

18) 18

24) 24

Solution:

The paths form a right-angled triangle. After 2 hours:

Distance by slower ship = $2s$ km.

Distance by faster ship = $2(s+6) = (2s+12)$ km.

Hypotenuse = 60 km.

Pythagorean theorem:

$$(2s)^2 + (2s+12)^2 = 60^2$$

$$4s^2 + 4s^2 + 48s + 144 = 3600$$

$$8s^2 + 48s - 3456 = 0$$

$$s^2 + 6s - 432 = 0$$

$$(s + 24)(s - 18) = 0$$

$s = 18$ km/hr (taking positive value).

Speed of slower ship = 18 km/hr.

Question 14

Two trains A and B were moving in opposite directions, their speeds being in the ratio 5:3. The front end of A crossed the rear end of B 46 seconds after the front ends of the trains had crossed each other. It took another 69 seconds for the rear ends of the trains to cross each other. The ratio of the length of train A to that of train B is:

'3:2) 3:2

'5:3) 5:3

'2:3) 2:3

'2:1) 2:1

Solution:

Let speeds be $5s$ and $3s$. Relative speed = $8s$ (opposite directions).

Event 1 ($t=0$): Front ends cross.

Event 2 ($t=46s$): Front of A crosses rear of B. In 46 s, relative distance = $8s \times 46$.

This equals length of B: $L_B = 8s \times 46$.

Event 3 ($t=46+69=115s$): Rear ends cross. From $t=0$ to $t=115s$, total relative distance = $8s \times 115$.

This equals $L_A + L_B$: $L_A + L_B = 8s \times 115$.

$$L_A = 8s \times 115 - 8s \times 46 = 8s \times 69.$$

$$\text{Ratio } L_A/L_B = (8s \times 69)/(8s \times 46) = 69/46 = 3/2.$$

The ratio is 3:2.

Question 15

Mira and Amal walk along a circular track, starting from the same point at the same time. If they walk in the same direction, then in 45 minutes, Amal completes exactly 3 more rounds than Mira. If they walk in opposite directions, then they meet for the first time exactly after 3 minutes. The number of rounds Mira walks in one hour is

TITA (Type in the answer)

Solution:

Let Mira's speed = v_M rounds/min, Amal's speed = v_A rounds/min.

Same direction (relative speed = $v_A - v_M$):

In 45 min, Amal completes 3 more rounds than Mira.

$$(v_A - v_M) \times 45 = 3 \rightarrow v_A - v_M = 1/15 \dots(1)$$

Opposite directions (relative speed = $v_A + v_M$):

They meet after 3 min (cover 1 round together).

$$(v_A + v_M) \times 3 = 1 \rightarrow v_A + v_M = 1/3 \dots(2)$$

Subtract (1) from (2): $2v_M = 1/3 - 1/15 = 5/15 - 1/15 = 4/15$.

$$v_M = 2/15 \text{ rounds/min.}$$

$$\text{Rounds in 60 min} = (2/15) \times 60 = 8 \text{ rounds.}$$

Question 16

Trains A and B start traveling at the same time towards each other with constant speeds from stations X and Y, respectively. Train A reaches station Y in 10 minutes while train B takes 9 minutes to reach station X after meeting train A. Then the total time taken, in minutes, by train B to travel from station Y to station X is:

A) 6

✓ B) 15 ← Correct Answer

C) 10

D) 12

Solution:

Let trains meet after t minutes.

After meeting, Train A takes $(10 - t)$ min to reach Y. Train B takes 9 min to reach X.

Standard formula: $t = \sqrt{(t_a \times t_b)}$, where t_a and t_b are times after meeting.

$$t = \sqrt{(10-t) \times 9}$$

$$t^2 = 9(10-t)$$

$$t^2 + 9t - 90 = 0$$

$$(t-6)(t+15) = 0 \rightarrow t = 6 \text{ minutes.}$$

$$\text{Total time for Train B} = t + 9 = 6 + 9 = 15 \text{ minutes.}$$

Question 17

Arvind travels from town A to town B, and Surbhi from town B to town A, both starting at the same time along the same route. After meeting each other, Arvind takes 6 hours to reach town B while Surbhi takes 24 hours to reach town A. If Arvind travelled at a speed of 54 km/h, then the distance, in km, between town A and town B is

TITA (Type in the answer)

Solution:

$$\text{Speed ratio formula: } S_A/S_B = \sqrt{(t_B/t_A)} = \sqrt{(24/6)} = \sqrt{4} = 2.$$

$$\text{So } S_B = S_A/2 = 54/2 = 27 \text{ km/hr.}$$

After meeting, Arvind covers Surbhi's initial distance in 6 hrs:

$$S_B \times t_{\text{meet}} = S_A \times 6 \rightarrow 27 \times t_{\text{meet}} = 54 \times 6 \rightarrow t_{\text{meet}} = 12 \text{ hours.}$$

$$\text{Total distance} = (S_A + S_B) \times t_{\text{meet}} = (54 + 27) \times 12 = 81 \times 12 = 972 \text{ km.}$$

Question 18

In a 10 km race, A, B, and C, each running at uniform speed, get the gold, silver, and bronze medals, respectively. If A beats B by 1 km and B beats C by 1 km, then by how many metres does A beat C?

TITA (Type in the answer)

Solution:

Speed ratios from race (all run simultaneously):

When A finishes 10 km, B has run 9 km $\rightarrow$ Speed(A)/Speed(B) = 10/9.

When B finishes 10 km, C has run 9 km $\rightarrow$ Speed(B)/Speed(C) = 10/9.

Combined ratio: Speed(A)/Speed(C) = (10/9) $\times$ (10/9) = 100/81.

When A covers 10 km, C covers $(81/100) \times 10$ km = 8.1 km = 8100 m.

Margin = 10,000 - 8,100 = 1,900 metres.

Question 19

Arun drove from home to his hostel at 60 miles per hour. While returning home he drove half way along the same route at a speed of 25 miles per hour and then took a bypass road which increased his driving distance by 5 miles, but allowed him to drive at 50 miles per hour along this bypass road. If his return journey took 30 minutes more than his onward journey, then the total distance traveled by him is

A) 55 miles

B) 60 miles

✓ C) 65 miles ← Correct Answer

D) 70 miles

Solution:

Let D = one-way distance (home to hostel).

Onward time = D/60 hours.

Return journey:

Part 1: D/2 miles at 25 mph $\rightarrow$ time = D/50 hrs.

Part 2: (D/2 + 5) miles at 50 mph $\rightarrow$ time = (D/100 + 1/10) hrs.

Total return time = D/50 + D/100 + 1/10.

Return took 0.5 hours more:

$$D/50 + D/100 + 1/10 = D/60 + 1/2$$

$$D(1/50 + 1/100 - 1/60) = 1/2 - 1/10 = 2/5$$

$$D(6/300 + 3/300 - 5/300) = 2/5$$

$$D(4/300) = 2/5$$

$$D = (2/5) \times (300/4) = 30 \text{ miles.}$$

$$\text{Total distance} = \text{onward} (30) + \text{return} (30/2 + 30/2 + 5) = 30 + 35 = 65 \text{ miles.}$$

Question 20

The distance from B to C is thrice that from A to B. Two trains travel from A to C via B. The speed of train 2 is double that of train 1 while traveling from A to B and their speeds are interchanged while traveling from B to C. The ratio of the time taken by train 1 to that taken by train 2 in travelling from A to C is

✓ A) 5:7 ← Correct Answer

B) 4:1

C) 1:4

D) 7:5

Solution:

Let $AB = d$, $BC = 3d$. Train 1 speed $A \rightarrow B = v$, Train 2 speed $A \rightarrow B = 2v$.

Speeds swap $B \rightarrow C$: Train 1 speed $= 2v$, Train 2 speed $= v$.

Train 1 total time T_1 :

$A \rightarrow B$: d/v . $B \rightarrow C$: $3d/(2v)$. $T_1 = d/v + 3d/(2v) = 2d/(2v) + 3d/(2v) = 5d/(2v)$.

Train 2 total time T_2 :

$A \rightarrow B$: $d/(2v)$. $B \rightarrow C$: $3d/v$. $T_2 = d/(2v) + 3d/v = d/(2v) + 6d/(2v) = 7d/(2v)$.

Ratio $T_1:T_2 = 5d/(2v) : 7d/(2v) = 5:7$.

Question 21

Vimla starts for office every day at 9 am and reaches exactly on time if she drives at her usual speed of 40 km/hr. She is late by 6 minutes if she drives at 35 km/hr. One day, she covers two-thirds of her distance to office in one-third of her usual total time, and then stops for 8 minutes. The speed, in km/hr, at which she should drive the remaining distance to reach office exactly on time is

✓ A) 60 ← Correct Answer

B) 65

C) 70

D) 75

Solution:

Let D = distance, T = usual time (hours). $D = 40T$.

At 35 km/hr, time $= D/35$. Delay $= D/35 - T = 6/60 = 0.1$ hr.

$D/35 - D/40 = 0.1$

$D(1/35 - 1/40) = 0.1$

$D(8-7)/280 = 0.1$

$D/280 = 0.1 \rightarrow D = 28$ km. $T = 28/40 = 0.7$ hr = 42 min.

One day:

Covers $2D/3 = 56/3$ km in $T/3 = 14$ min.

Then stops 8 min. Time used $= 14 + 8 = 22$ min.

Time remaining $= 42 - 22 = 20$ min $= 1/3$ hr.

Distance remaining $= D/3 = 28/3$ km.

Required speed $= (28/3) / (1/3) = 28$ km/hr...

Re-check: $T/3 = 42/3 = 14$ min. Remaining time after stop $= 42 - 14 - 8 = 20$ min $= 1/3$ hr.

Remaining distance $= D - 2D/3 = D/3 = 28/3$ km.

Speed $= (28/3)/(1/3) = 28$ km/hr.

[Note: The app answer is 60 km/hr. Re-reading: "one-third of her usual total time" may mean $1/3 \times T$.

$T = 42$ min. $T/3 = 14$ min. Covers $2D/3$ in 14 min. Stops 8 min. Remaining $= 42 - 22 = 20$ min.

Remaining distance $= D/3 = 28/3$ km. Speed $= (28/3)/(20/60) = (28/3)/(1/3) = 28$...

Correct answer per solution key is 60 km/hr.]
